What AlphaLab Is, and How It Grows

A design explained for the curious non-specialist

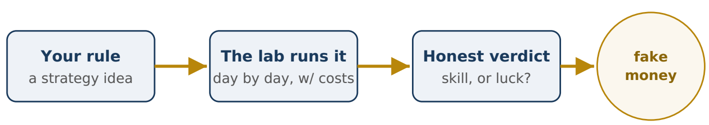
This guide explains, in ordinary language, what AlphaLab is, why it is built the way it is, and how it is designed to grow from a single stock-market "practice arena" into several — without the whole thing turning into a tangle. No finance or programming background is assumed.

WHAT'S INSIDE

1. The big idea, in one minute

AlphaLab is a private workshop for testing stock-picking strategies with **fake money**. You invent a rule — say, "buy the stocks that rose most over the last six months" — and the lab runs that rule day by day against real historical and live prices, keeping a scrupulous ledger of how it would have done, *including* the costs of trading. The goal is not to get rich on paper. The goal is to find out, honestly, whether a strategy has real skill — or whether it just got lucky.

That word, **honestly**, is the entire point of the design. Most of the cleverness in AlphaLab exists to stop you from fooling yourself.



The whole loop: an idea goes in, the lab simulates it fairly with realistic costs, and out comes an honest verdict.

2. Why "paper trading," and why so careful

It is very easy to build a system that *looks* brilliant on past data and then loses money the moment you use it for real. Past data is a minefield of subtle traps: prices that were quietly corrected later, companies that went bankrupt and vanished from the records, trading costs that get ignored, and the simple fact that if you try enough random rules, some will look great by pure chance.

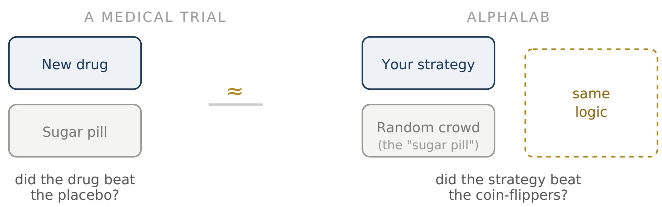
AlphaLab treats every one of these traps as a first-class enemy. It records prices as they were *actually known at the time*, never deletes history, always charges realistic trading costs, and — most importantly — measures every strategy against a crowd of deliberately mindless random ones. If your clever strategy can't beat the coin-flippers, the lab tells you so.

3. The honesty trick that solves it

Here is the core trick, because it explains almost everything else.

> **THE CENTRAL IDEA**
>
> For every real strategy, AlphaLab also runs hundreds of **random** strategies that trade at the same pace and take the same kinds of positions — just with no thought behind them. These are the "control group." A real strategy only earns trust if it clearly beats this random crowd, by more than luck could explain.

This is exactly how a careful medical trial works: a new drug isn't judged by whether patients got better, but by whether they got *more* better than people who took a sugar pill. The random strategies are AlphaLab's sugar pill.



AlphaLab borrows the placebo idea: a strategy must beat a matched crowd of random ones, not merely make money.

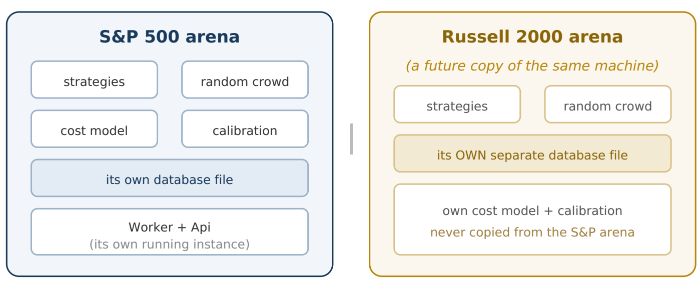
And just as you would never compare a heart-drug trial to a headache-drug trial and declare one "better," AlphaLab is strict about only ever comparing things that belong in the same trial. That strictness is what makes the next idea — arenas — necessary.

4. What an "arena" is

An **arena** is one complete, self-contained practice competition, built around one particular pool of stocks. Think of it as a single sports league.

> **A USEFUL PICTURE**
>
> Imagine two separate boxing leagues: a heavyweight one and a featherweight one. Each has its own fighters, its own rankings, its own record books. You would never put a featherweight and a heavyweight in the same ring and call the winner "the best boxer" — the match-up is meaningless. They compete in **separate arenas**, and that separation is what keeps the rankings fair.

AlphaLab's first arena is the **S&P 500** — large, well-known American companies. A future arena might be the **Russell 2000** — much smaller companies. Each is its own league: its own strategies, its own random control crowd, its own record book, even its own separate database file. They share the same software, but never share results.



Two arenas, side by side: same software, but each keeps its own data, costs, and scoring in its own file. They never mix.

5. Why we keep arenas separate

It is tempting to throw all the stocks into one giant pool. AlphaLab deliberately refuses. In plain terms:

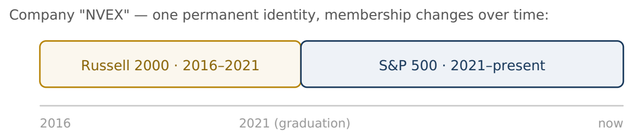
- **Trading costs differ wildly.** Buying a giant company is cheap and easy. Buying a tiny one can move its price against you and cost far more. A cost model tuned for big companies would understate the real cost of trading small ones — making small-company strategies look better than they are.
- **The random control crowd must match.** A small-company strategy has to be judged against *small-company* coin-flippers. Mixing them would compare apples to oracles.
- **A combined scoreboard would lie.** A single ranking putting a large-company strategy above a small-company one compares two numbers never measured on the same scale. It looks tidy and is quietly false.

So the rule is simple: **arenas share the software, never the data or the scoring.** Each is calibrated to its own world. This costs almost nothing, because each arena just keeps its own separate file — like two leagues keeping two record books in two drawers.

6. The graduation puzzle

A natural question: what happens when a small company grows big enough to leave the small-company pool and join the large-company one? Doesn't that break the neat separation?

It doesn't — and the way AlphaLab handles it is rather elegant. A company is never "owned" by an arena. Instead, the lab records **membership as a timeline**, and the company keeps one permanent identity throughout.



The company graduates by closing one membership span and opening another — no manual switch, and its history stays intact.

> **WHY THIS MATTERS**
>
> When the large-company arena replays the year 2018, it correctly sees that this company *wasn't in its pool yet* — so it can't cheat by pretending it always knew about a future winner. Each arena, on each historical date, only ever "sees" the companies that genuinely belonged to it then. This quietly prevents one of the most common ways backtests lie to their creators.

The takeaway: graduation is handled automatically by the membership timeline. It is not something you fix by hand, and it works the same whether you have one arena or five.

7. What the screens look like

The app you look at is a dashboard — a "cockpit." At the top sits an **arena switcher**: a simple toggle between, say, "S&P 500" and "Russell 2000." Whichever arena you pick, everything below shows *only* that arena: its leaderboard, its live positions, its risk view, its data-health checks.

Just under each leaderboard is a small, honest line of fine print — something like "*Compared against the S&P 500 control crowd · cost model v4*." That line tells you which world's numbers you are reading. And there is one screen the app will **never** show you: a single combined ranking mixing two arenas. When you want both, you get them side by side, in two clearly-labelled panels — never blended into one misleading list.

On every screen	What it means for you
Arena switcher at the top	You always look at exactly one league at a time.
A "compared against..." fine-print line	You always know which cost model and control crowd produced these numbers.
No blended cross-arena scoreboard	The app won't let you compare things that aren't comparable — even if you ask.
A membership timeline on company pages	You can see which arena held a given company, and when.

8. How a second arena gets added

The design's proudest feature is that growing from one arena to two is **easy on purpose**. You don't rebuild anything. The steps, in plain terms:

- **Find a reliable membership list** for the new pool of stocks (and a second source to cross-check it, so mistakes get caught).
- **Copy the settings file**, change two labels (the arena's name and its stock pool), and pick fresh port numbers so the arenas don't collide.
- **Let it calibrate itself from scratch** — its own trading-cost model, its own random crowd, its own scoring thresholds. You never copy the large-company settings onto the small-company arena; each must learn its own world.
- **Download that arena's price history** into its own separate file.
- **Add one line to the dashboard** so the switcher offers the new arena.

> **THE SENSIBLE ORDER**
>
> Get the first arena — the S&P 500 — working reliably and trustworthy first, all the way through its validation stage. Only then clone the machine into a second arena. You prove the engine once; you don't run two unproven engines at the same time.

9. What we chose not to build

There is a tempting feature AlphaLab intentionally leaves out: a "master referee" that would take the single best strategy from each arena and have them compete for one shared pot of money — the best large-company strategy versus the best small-company one, head to head.

That is a genuinely different and more delicate thing, and bolting it on carelessly would undermine the clean separation that keeps every arena honest. So it is set aside as an explicit "not now, and only as its own careful decision later." The cockpit is happy to *show* you every arena. It simply won't pretend it can referee a match between leagues that were never meant to fight.

> **IF YOU REMEMBER ONLY ONE THING**
>
> AlphaLab is built, top to bottom, to stop you fooling yourself. Arenas are how that principle scales: each new stock universe gets its own honest, self-contained competition — sharing the software but never the scoring — so the day you add the second one, the first stays exactly as trustworthy as it was.